

Marine Antifouling Coatings Based on Durable Bottlebrush Polymers

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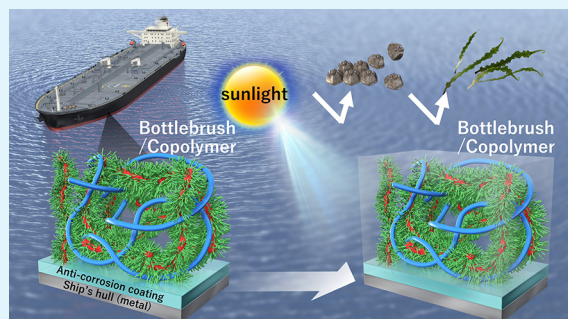
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Supporting Information

ABSTRACT: We report a next-generation, biocide-free, and durable marine antifouling coating technology. To achieve this, we combined two different polymers previously developed by us. First, we synthesized well-defined 2-hydroxypropyl acrylamide (HPA) based bottlebrush polymers with concentrated polymer brush (CPB) structures, which exhibit excellent bioinertness, and second, we synthesized photoreactive copolymers of 2-hydroxypropyl acrylamide (HPA) and *N*-benzophenone acrylamide (BPA), which can be cross-linked by exposure to sunlight for 30 min. Simply mixing the bottlebrush polymers with the photoreactive copolymers and applying these as a coating provided a scalable method for achieving effective antifouling properties in one step on a broad range of polymer substrate materials. The resistance of bottlebrushes against acid and base hydrolysis was demonstrated in accelerated degradation experiments at 80 °C, and the coating thickness was found to be stable after 3 months of incubation in sea water. Optimized coatings prevented cypris larva attachment for up to 9 days and prevented the settling of marine organisms in the sea for up to 73 days. Due to the ease of application, long-term durability, and effective antifouling performance, our bottlebrush coating technology is expected to be exploited in biocide-free marine paints.

KEYWORDS: concentrated polymer brush, bottlebrush polymer, photoreactive polymer, marine biofouling, 2-hydroxypropyl acrylamide, *N*-benzophenone acrylamide



INTRODUCTION

Marine biofouling is an exceptionally important problem for many applications where material surfaces come into contact with the marine environment. The most prominent example is shipping vessels due to the fact that global logistics by maritime navigation is indispensable for the world economy. While it is a more cost-effective and environmentally friendly transportation compared to airplanes and other transportation methods, it is essential to reduce greenhouse gas emissions from international shipping, which are contributing to the recent acceleration of global warming and climate change.^{1–7} Therefore, the International Maritime Organization (IMO) has recently adopted new international rules and set an initial strategy (as the short-term goal) to reduce carbon dioxide (CO₂) emissions by 40% at least by 2030 compared to 2008.¹ To achieve this goal, a multitude of new and improved technologies are required, including construction of more fuel-efficient ships. Here, the key to improving energy efficiency is to reduce fluid frictional resistance, which accounts for 60–80% of the water resistance experienced by a ship's hull. It is generally known that marine biofouling spontaneously accumulating on the ship's hull is a major contributor to frictional resistance. Surprisingly, even light fouling with slimes can reduce the speed of a ship by 10–20% due to friction.²

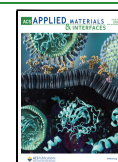
Therefore, to prevent marine organisms from colonizing submerged surfaces and to reduce frictional resistance, ship hulls are commonly coated with an antifouling paint.^{3–7}

Major antifouling paints on the market contain cuprous oxide as a biocide. Although the products are effective at preventing biofouling and the environmental safety is considered, the possibility cannot be denied that long-term leaching and the accumulation of cuprous oxide in the ocean may have adverse environmental effects, similar to tributyl tin biocides that were ruled out for this application due to their long-term damage to the environment. Therefore, many efforts have been made to develop biocide-free, nonreleasing antifouling coating technologies. Novel approaches that are being explored include bioinspired engineered topographies, polymer brushes, nanocomposites, etc.^{3–7} While these approaches have demonstrated the successful prevention of biofouling in fundamental studies, their scale-up often is

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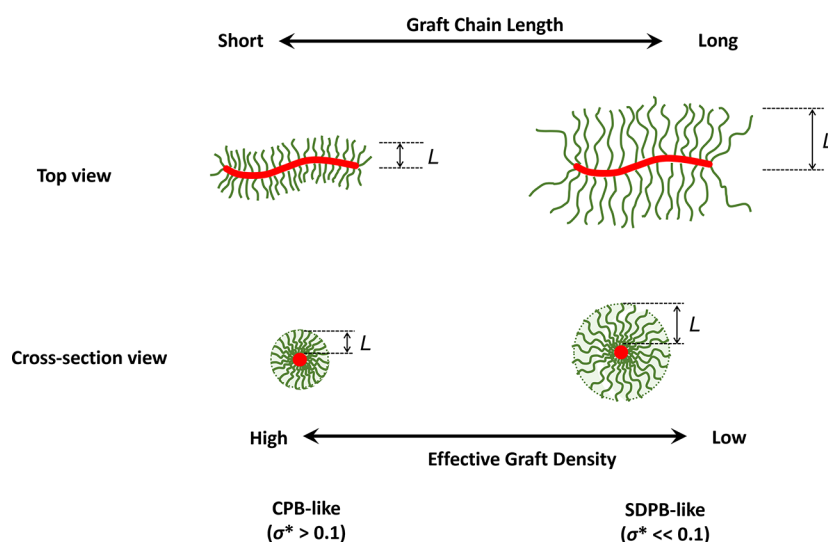


Figure 1. Schematic illustration of the columnar model for bottlebrush polymers. If the same number of polymer chains is grafted on the stem polymer, then a higher effective graft density (as a function of σ^*) is achieved with the bottlebrush polymer having shorter graft chains.

difficult and their performance in field tests is often unsatisfactory when compared to commercial paints containing biocides. In addition, durability is also an important issue to consider. Since commercial paints can prevent biofouling for more than 5 years, alternative biocide-free paints should provide the same performance. In summary, novel coating technologies based on the prevention of biofouling require an easy-to-apply coating method, sufficient physical stability for several years, and highly effective antifouling properties.

In the context of biofouling, high density polymer brushes prepared by surface-initiated living radical polymerization (SI-LRP) have been reported to provide outstanding performance.^{8–10} In particular, neutral hydrophilic polymers obtained from poly(ethylene glycol) methyl ether methacrylate (PEGMA), 2-hydroxyethyl methacrylate (HEMA), and 2-hydroxypropyl acrylamide (HPA), as well as zwitterionic polymers obtained from 2-methacryloyloxyethyl phosphorylcholine (MPC), carboxybetaine [1-carboxy-*N,N*-dimethyl-*N*-(2'-methacryloyloxyethyl)methanaminium inner salt] (CMB), and sulfobetaine methacrylamide (SBAm), are often used in biomedical applications because of their high biocompatibility.^{8–10} As their low-biofouling properties are expected to be applicable also in the marine environment, multiple studies have also assessed these polymer brushes in regard to their ability to prevent the settlement of green algae and barnacle cyprids.^{11,12} While these studies have shown promising results in regard to preventing the early stages of marine biofouling, these coatings cannot be used on a marine vessel. This is due to the fact that it is impossible to directly graft polymer brushes onto a large ship hull by SI-LRP due to the multistep process requiring an inert gas atmosphere and the removal of residual monomer.

It is known that high density polymer brushes, so-called concentrated polymer brushes (CPBs), obtainable by SI-LRP show unique structures and properties that have never been achieved with semi-dilute polymer brushes (SDPBs).^{13–17} For example, in a good solvent, polymer chains of the CPB are stretched to 80–90% of their full lengths.¹⁴ Reflecting the high-extension structure, the swollen CPBs exhibit strong repulsion against compression,¹⁵ super-lubrication,^{16,17} and a size-exclusion effect.^{18,19} Based on the scaling theory, CPBs or

SDPBs can be classified by their dimensionless graft densities (σ^*): When $\sigma^* > 0.1$, a polymer brush is classified as CPB.¹³ Thus, focusing on the graft density, we newly designed bottlebrushes that have a CPB structure (Figure 1) and cross-linked the bottlebrushes by radical coupling at the chain ends, developing a novel bioinert coating using the CPB-type bottlebrushes instead of direct CPB surface grafting.²⁰ In the previous work, we confirmed that the CPB-type bottlebrush coatings significantly reduced protein adsorption as well as cell and bacterial adhesion when compared with SDPB-type polymers.²⁰ The antifouling properties achieved were almost the same as those of CPBs directly grafted on silicon wafers.^{21–26} Since the CPB-type bottlebrushes are relatively easy to coat on material surfaces and have promising bioinert properties, we planned to expand this bottlebrush coating to marine antibiofouling applications.

Herein, we focused our attention on two issues: Ease of coating application and long-term stability against hydrolysis (Figure 2). Recently, we developed low biofouling coatings that can be applied in a single step using photoreactive

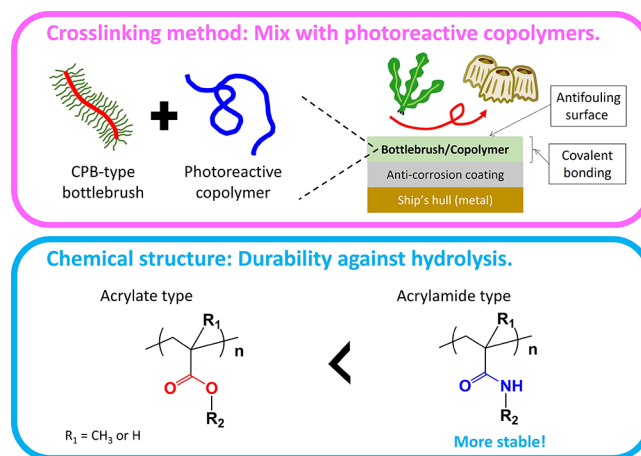


Figure 2. Strategy of designing bottlebrush coatings utilizing photoreactive copolymers and the preferable chemical structure for monomers.

copolymers of HPA (low biofouling component) and *N*-benzophenone acrylamide (BPA) (photoreactive component).²⁷ The copolymers can be easily coated and cross-linked by UV irradiation while also being covalently bound to a broad range of polymeric substrates even in the absence of specific functional groups. Here, the cross-linking density can be controlled by varying the molar ratio of BPA in the copolymer. Thus, we planned to utilize the copolymers to covalently immobilize bottlebrushes on the surfaces. Typically, the ship hull is first coated with antirust resin and then overlaid with antifouling paints. Therefore, the photoreactive copolymers provide an opportunity to covalently immobilize the bottlebrushes on a ship hull (the resin surface). Moreover, to achieve long-term stability against hydrolysis, we selected acrylamide monomers for the synthesis of bottlebrushes and photoreactive copolymers, as it is well known that amides are much more stable in water compared to esters because of the difference of polarization between amide bonds and ester bonds.²⁸

In this study, we first confirmed that bottlebrushes composed of HPA (amide) were more stable against alkaline and acid hydrolysis compared with those composed of PEGMA (ester). Secondly, we optimized the bottlebrush coatings by controlling the BPA molar ratio in the copolymers and by changing the mixing ratio of the copolymers to the bottlebrushes. We also demonstrated that the photoreactive copolymers can be easily cross-linked by exposure to sunlight, an aspect that is important when coating a large area such as a ship hull. Moreover, we demonstrated long-term stability (2 months) of the coatings in artificial sea water by monitoring thickness decrease and evaluated the antifouling properties using mammalian cell attachment experiments and barnacle cypris larva settlement tests on the coatings. Finally, we carried out a field test by immersing coatings in the ocean for a period of up to 2 months. The results confirmed that our coatings provide excellent antifouling properties in the marine environment while also offering long-term stability and a scalable coating application process. Thus, our bottlebrush coating technology is expected to be exploited in alternative biocide-free marine paints.

MATERIALS AND METHODS

Materials. 4-Chloromethylstyrene (>99.5%, TCI Co. Ltd., Tokyo, Japan), 4-cyano-4-[(dodecylsulfanylthiocarbonyl)sulfanyl]pentanoic acid (>97.0%, Wako Pure Chemical Industries, Osaka, Japan), methanol (99.8%, Wako Pure Chemical Industries, Osaka, Japan), 2,2'-azobis(isobutyronitrile) (AIBN) (>98%, Wako Pure Chemical Industries, Osaka, Japan), tris[2-(dimethylamino)ethyl]amine (Me6TREN) (>98.0%, TCI Co. Ltd., Tokyo, Japan), copper(I)-chloride (Cu(I)Cl) (>97.0%, Wako Pure Chemical Industries, Osaka, Japan), *N,N*-dimethylformamide (DMF) (>99.5%, Wako Pure Chemical Industries, Osaka, Japan), 3-aminopropyltriethoxysilane (AMP) (>98.0%, TCI Co. Ltd., Tokyo, Japan), and artificial sea salt (LIVESea Salt) (DELPHIS Co. Ltd., Hyogo, Japan) were used as received. Poly(ethylene glycol) methyl ether methacrylate (PEGMA, average molecular weight ~500, 99%, Aldrich) was purified by passing through a column of basic alumina. 2-Hydroxypropyl acrylamide (HPA) and *N*-benzophenone acrylamide (BPA) were synthesized according to our previously described methods.²⁷ 2-(2-Bromoisobutyryloxy)ethyl methacrylate (BIEM) and polyBIEM (PBIEM) were synthesized according to the literature.²⁹ Polyethylene terephthalate (PET) films and polyvinyl chloride (PVC) films (thickness = ~200 μ m) were purchased from AS ONE Corp. (Osaka, Japan).

Synthesis of Poly[4-(chloromethyl)styrene] (PST(Cl)), a Macroinitiator Made by RAFT Polymerization. 4-Chloromethylstyrene (4.0 g, 2.6×10^{-2} mol), BM1432 (21 mg, 5.24×10^{-5} mol), and AIBN (5.2 mg, 2.1×10^{-5} mol) were prepared in a nitrogen purged glove box and transferred into a Schlenk tube. Then the solution was mixed at 70 °C for 7 h. After the polymerization, the monomer conversion was determined by ¹H NMR analysis (ESC-400 spectrometer, JEOL Ltd., Tokyo, Japan). The number-averaged molecular weight (M_n) and polydispersity (M_w/M_n) were measured by polystyrene standard (PST) calibrated-gel permeation chromatography (GPC). The polymer was purified by reprecipitation with methanol to remove monomers and other residues.

Synthesis of Bottlebrushes: ATRP of HPA from PST(Cl). A DMF/MeOH (2/1 vol/vol %) solution of PST(Cl) (40 – x mM), HPA (2.0 M), Cu(I)Cl (40 mM), and Me6TREN (80 mM) with and without benzyl chloride (free initiator; $x = 8$ or 0 mM, respectively) was prepared in a nitrogen-purged glove box and transferred to a Schlenk flask. Next, the solution was stirred at 30 °C for the prescribed time. After polymerization, the number-averaged molecular weight (M_n), the weight-averaged molecular weight (M_w), and the polydispersity (M_w/M_n) were determined by PST-calibrated GPC. The conversion rate of PHPA was determined by ¹H NMR analysis. The bottlebrush polymers that were obtained by polymerization without free initiators were reprecipitated with acetone and purified by dialysis (MWCO = 3.5 kDa, Spectra Pore 3). Subsequently, the aqueous solution was freeze-dried to obtain the bottlebrush polymers (yellowish white powder).

Calculation of Graft Density. Figure 1 shows a schematic illustration of the columnar model for bottlebrush polymers. The effective graft density (σ_{eff}) of the bottlebrushes, that is, whether they have a CPB- or SDPB-type structure ($\sigma^* > 0.1$ or $\sigma^* \ll 0.1$), is determined by their graft chain length ($L = 0.25 \times \text{DP}_{n,\text{graft}}$). In the case of poly(acrylamide)-type bottlebrushes, these can be calculated as follows:

$$\sigma_{\text{eff}} = \frac{1}{\{2\pi(\text{DP}_{n,\text{graft}} \cdot 0.25 + r)\} \left(\frac{0.25}{x}\right)} \quad (1)$$

$$\sigma^* = a^2 \sigma_{\text{eff}} \quad (2)$$

where $\text{DP}_{n,\text{graft}}$ is the number-averaged degree of polymerization of the graft chains, x is the number of graft chains per unit stem polymer (initiation efficiency of the macroinitiator) ($x = 1$ for the PST(Cl) against HPA), r is the radius of the backbone polymer (0.6 nm for the PST(Cl)), and a^2 is the cross-sectional area of the monomer (0.9 nm² for HPA). The factor of 0.25 is the full length of the repeating unit of the polymethacrylate in question (C-C-C, trans zigzag) so that $2\pi(\text{DP}_{n,\text{graft}} \cdot 0.25 + r)$ is the circumference of the cross-section of a bottlebrush and $0.25/x$ is the distance between the neighboring graft chains. Assuming that the number of graft chains per unit stem polymer is constant, the effective graft density at the outermost periphery decreases with an increase in the chain length. Based on the scaling theory, the bottlebrush polymers can be categorized into CPB or SDPB by σ^* .¹³

Stability of Bottlebrushes against Hydrolysis. Polymethacrylate analogous bottlebrushes, PPEGMA-*g*-PBIEM, were prepared according to our previous report.²⁰ The PHPA-*g*-PST(Cl) and PPEGMA-*g*-PBIEM were labeled as bottlebrush (HPA) and bottlebrush (PEGMA) (Figure S1). The acidic aqueous solution (pH = 1) was prepared at a concentration of 1 M HCl in D₂O, and the alkali aqueous solution (pH = 14) was prepared at a concentration of 1 M NaOH in D₂O. The bottlebrushes (20 mg) were dissolved in the aqueous solutions (0.8 g), then transferred into NMR tubes, and heated in an oil bath at 80 °C. At the prescribed time, the NMR tubes were taken out of the oil bath and immediately cooled down in an ice bath to ambient temperature. Then the polymers were analyzed by ¹H NMR at ambient temperature. After the NMR measurement for 6 h, the polymer solutions were immediately neutralized and measured by GPC.

Random Copolymerization of HPA and BPA: Photoreactive Copolymer (Poly(HPA-*r*-BPA)). Random copolymers of HPA and BPA were prepared according to our previous report.²⁷ The feed molar ratio of the BPA monomers was varied from 1 to 10 mol %. As an example, the synthesis of a copolymer with a molar ratio of HPA/BPA = 99/1 is described here briefly. A DMF solution containing HPA (2.3 M), BPA (23 mM), and AIBN (2.32 mM in water) was purged with argon gas, and the solution was magnetically stirred at 70 °C for 1.5 h. After the polymerization, the monomer conversion was determined by ¹H NMR. The number-averaged molecular weight (M_n), the weight-averaged molecular weight (M_w), and the polydispersity (M_w/M_n) were determined by gel permeation chromatography (GPC). The copolymers were reprecipitated with acetone and subsequently purified by dialysis (MWCO = 3.5 kDa, Spectra Pore 3). The aqueous solution was then freeze-dried to obtain the pure copolymers (white powder).

Preparation of Cross-Linked Coatings. An ethanol solution containing 3% (w/w) of the random copolymers and an ethanol solution containing 3% (w/w) of the bottlebrush were mixed at 1/1 or 1/2 (v/v). Then, the mixed solutions were coated on PET or PVC films using a spin coater (MS-A-100) (MIKASA Co., Ltd., Tokyo, Japan) at 3000 rpm for 60 s. The coated films were subsequently cross-linked by UV irradiation (UV-A: 170 mW/cm²) (HB100A-1, AS ONE Co., Ltd., Osaka, Japan) for 4 min. The spin-coated films were also cross-linked by a ceiling lamp (fluorescent tube lamp) (~150 μW/cm²) or sunlight (~6 mW/cm²) for the prescribed time.

X-ray Photoelectron Spectroscopy (XPS) Analysis. XPS analysis was performed using an AXIS Nova spectrometer (Kratos Analytical Inc., Manchester, UK) with a monochromated Al K α source at a power of 180 W (15 kV $\times$ 12 mA) and a hemispherical analyzer operating in the fixed analyzer transmission mode. The total pressure in the main vacuum chamber during analysis was typically between 10⁻⁹ and 10⁻⁸ mbar. Survey spectra were acquired at a pass energy of 160 eV and step size 0.5 eV. To obtain more detailed information about chemical structure, oxidation states, etc., high-resolution spectra were recorded from individual peaks at a pass energy 20 eV and step size 0.1 eV, typically yielding a FWHM for the ester peak in PET of less than 0.85 eV during performance tests. Each specimen was analyzed at an emission angle of 0° as measured from the surface normal. Assuming typical values for the electron attenuation length of relevant photoelectrons, the XPS analysis depth (from which 95% of the detected signal originates) ranges between 5 and 10 nm for a flat surface.

Data processing was performed using CasaXPS processing software version 2.3.21 (Casa Software Ltd., Teignmouth, UK). All elements present were identified from survey spectra. The atomic concentrations of the detected elements were calculated using integral peak intensities and the sensitivity factors supplied by the manufacturer. Binding energies were referenced to the C 1s peak at 284.8 eV (aromatic hydrocarbon). Peak fitting of C 1s spectra was performed using a three-parameter universal Tougaard background and components having a generalized Voigt line shape (convolution of Lorentzian and Gaussian function).

The accuracy associated with quantitative XPS is ca. 10–15%. Precision (i.e., reproducibility) depends on the signal/noise ratio but is usually much better than 5%. The latter is relevant when comparing similar samples.

Gel Permeation Chromatography (GPC) Analysis. The GPC analysis of copolymers produced in this study was performed using an HLC-8220 instrument (TOSOH Corporation, Tokyo, Japan) equipped with a Shodex LF-804 column (40 °C) and 10 mM LiCl in DMF as the eluent with a flow rate of 0.8 mL/min. Polystyrene standards (PStQuick, TOSOH Corporation) were used as the reference.

Thickness Measurements. The thickness of coating layers was investigated using an ellipsometer (MAXY-102SM (SP)). The instrument was used in conjunction with a He–Ne laser (λ = 632.8 nm) at an angle of incidence of 70°. Measurements were averaged over at least nine points on the surface of each sample.

Contact Angle Measurements. Static water contact angle measurements were carried out using a contact angle goniometer (VCA-Optima XE, AST Products Inc., Billerica, MA, USA). A water droplet was placed on the substrate in each case, and the angle was measured after 20 s. The averaged data were obtained from 10 separate measurements.

Stability of the Polymer Coatings in Seawater. Artificial seawater was prepared with the artificial sea salt (LIVESea Salt) (Delphis Co., Ltd.) according to the instruction manual. The PET films with polymer coatings and bare PET films (round shape, radius = 0.79 cm) were placed in 24-well culture plates, and 1 mL of artificial seawater was added. The multiwell plates were then stored at 25 °C. Every 2 or 3 days, the artificial seawater was exchanged. At 1, 2, and 3 months, the films were taken out of the artificial seawater and washed by dipping in Milli-Q water for 18 h. After drying, the dry thickness of the polymer coatings was measured using an ellipsometer.

Cell Culture Experiments. The mouse fibroblast cell line (L929) (RCB1422) used in cell culture experiments was purchased from RIKEN Bioresource Center (Tsukuba, Japan). The L929 cells were maintained in Dulbecco's modified Eagle medium (DMEM) (Aldrich) supplemented with 10% fetal bovine serum (FBS), 100 U mL⁻¹ penicillin, and 100 μg mL⁻¹ streptomycin at 37 °C in a humidified atmosphere containing 5% CO₂ for 4–5 days until achieving a confluence of 70–80%. The cells were removed from the culture flasks using 0.05% trypsin–0.53 mM EDTA and resuspended at a concentration of 1.0 $\times$ 10⁵ cells mL⁻¹ in DMEM.

The PET films with polymer coatings as well as untreated PET films (5 $\times$ 5 cm²) were placed in cell culture dishes (diameter = 100 mm), and 15 mL of artificial seawater was added. The dishes were stored at 25 °C. Every 2 or 3 days, the seawater was exchanged. At 1 and 2 months, the films were taken out from the seawater and washed by dipping in Milli-Q water for 18 h. After drying, the films were cut into small pieces (1 $\times$ 1 cm²) and placed into 24-well plates. Before cell seeding, the films were sterilized with 70% ethanol. Cells were then seeded on the surfaces at a concentration of 5.0 $\times$ 10⁴ cells cm⁻² and cultured for 24 h at 37 °C in a humidified atmosphere containing 5% CO₂. The films were gently washed three times with phosphate-buffered saline (PBS) and placed in 4% paraformaldehyde overnight at 4 °C to fix the adherent cells on the surfaces. The adherent cells were stained with Alexa Fluoro 555 (Invitrogen, CA, USA) and mounted with a Prolong Gold antifade reagent with 4',6-diamidino-2-phenylindole (DAPI) (Invitrogen) to stain the nuclei in blue and the actin filaments in red. The adherent cells were then observed by fluorescence microscopy. We prepared three samples for each surface type, and at least seven spots were observed on each surface.

Preparation of Barnacle Cypris Larvae. *Balanus amphitrite* were obtained from the coast of Himeji in the Seto Island Sea (Hyogo, Japan). They were kept at 23 $\pm$ 1 °C in a circulation aquarium and fed brine shrimps. Nauplius larvae hatched from the *Balanus amphitrite* were transferred to a glass beaker with filtered natural seawater. The nauplius larvae were fed *Chaetoceros calcitrans* until they became cypris larvae. The cypris larvae were stored at 6 °C under dark conditions until use. Before a settlement test, the seawater containing the cypris larvae was increased to 23 °C. The actively swimming larvae were collected and used for the settlement test.

Cypris Larva Settlement Test. PVC films with polymer coatings as well as untreated PVC films (40 $\times$ 40 mm) were used for the cypris larva settlement test. Each sample was placed in a polypropylene container (W 111 $\times$ L 81 $\times$ D 46 mm), and 100 mL of seawater containing the cypris larvae (100 individuals/container) was transferred onto the sample (Figure S6). The temperature was maintained at 23 °C, and the container was placed under a light during daytime and kept in the dark during night time. Cypris larvae that actively walked along the substrates with antennule attachment were termed as "searching", and those in the temporary adhesion stage were termed as being in the "attachment" phase (Scheme S2, 2. Attachment and searching). The cypris larvae that were searching and attached on the substrate as well as those undergoing metamorphosis into juveniles (Scheme S2, 3. Metamorphosis) were counted as "settled". The experimental run number was two.

Static Immersion Test in the Sea (Field Test). The cross-linked BPA 5 mol % coatings with or without bottlebrush prepared on a PET film ($5 \times 5 \text{ cm}^2$) as well as an untreated PET film were immobilized on the test panel using epoxy resin. The panel surfaces not covered by the PET films were covered with antifouling paint. The test panels were immersed at a depth of 1.5 m in the sea at Sazaurara, Mie, Japan ($34^\circ 19' 07.4'' \text{ N } 136^\circ 40' 12.6'' \text{ E}$) for 2.4 months (from August to October; with an average water temperature of approx. 25°C). The panels were taken out of the sea and washed with seawater. Pictures of the surfaces were taken to enable the observation of settled marine organisms and the fouled area.

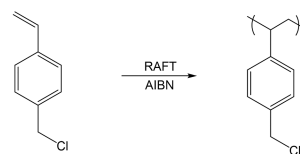
RESULT AND DISCUSSION

Preparation of Bottlebrushes with a Concentrated Polymer Architecture. All industry sectors have a need for the effective prevention of biofouling, ranging from the health care to the blue economy sector. This has resulted in the development of a broad range of technologies and here in particular coating technologies. Out of the range of technologies on offer, those that focus on the passive prevention of biointerfacial interactions at the molecular level have been driven by the health care sector due to the need for nontoxic, biocompatible technologies. Here, high-density graft polymer coatings have for a long time been considered as the gold standard,^{8–10,18–26,30} but cross-linked polymer coatings that can be applied in a single step have more recently taken this place due to their effectiveness and suitability in regard to the translation into manufacturing.^{27,31–33} Previously, we successfully synthesized bottlebrush polymers that have a CPB structure and confirmed that coatings containing these bottlebrush polymers showed excellent capability in regard to the prevention of biofouling compared with polymers having a SDPB structure.²⁰ However, in the research, poly(meth)acrylate polymers were employed for both the macroinitiator (PBIEM) and bottlebrush (PEGMA) components (see Figure S1). As these ester moieties are susceptible to hydrolysis, we have chosen to use polymers having a much more stable chemistry, consisting of amide moieties (i.e., polyacrylamides) or polystyrene moieties in this study. More specifically, to enhance the stability of bottlebrushes against degradation by aqueous media such as seawater and biological fluids, we used PST(Cl) as the macroinitiator and HPA as the graft polymer.

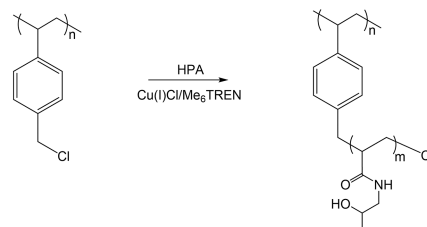
First, PST(Cl), a macroinitiator, was prepared by RAFT polymerization (Scheme 1a). Table 1 shows characteristics of the macroinitiator. From the conversion and the initial monomer ratio to RAFT agent, we obtained PST(Cl) with an approximate number-averaged degree of polymerization (DP_n) = 150. Then, we optimized the ATRP condition for the HPA monomer using PST(Cl) (Scheme 1b). As shown in Figure 1, the graft chain length ($L = 0.25 \times \text{DP}_n$) of the columnar bottlebrush affects the effective graft density (σ_{eff}) of the bottlebrushes, namely, whether the bottlebrush forms a CPB or SDPB structure can be determined by their graft chain length (L). Herein, we added benzyl chloride, a free initiator in the system ($[\text{benzyl chloride}]_0/[\text{PST(Cl)}]_0 = 1/10$). It is known that polymers grown in solutions from these initiators are good indicators regarding the M_n and M_w/M_n for graft polymers.¹³ Table S1 shows the characteristics of bottlebrushes and free polymers. We confirmed that the initiation efficiency of PST(Cl) (x) was almost 100% using GPC peak areas, as described in our previous report.²⁰ From the values of σ^* , the bottlebrushes were confirmed as CPBs. Thus, we adopted 2 h as the polymerization time to obtain CPB-type bottlebrushes. To obtain pure bottlebrushes (without free polymers), we

Scheme 1. Synthesis of the (a) Macroinitiator (PST(Cl)) and (b) Bottlebrush Polymer (HPA)

(a) Synthesis of macro-initiator



(b) Synthesis of bottlebrush



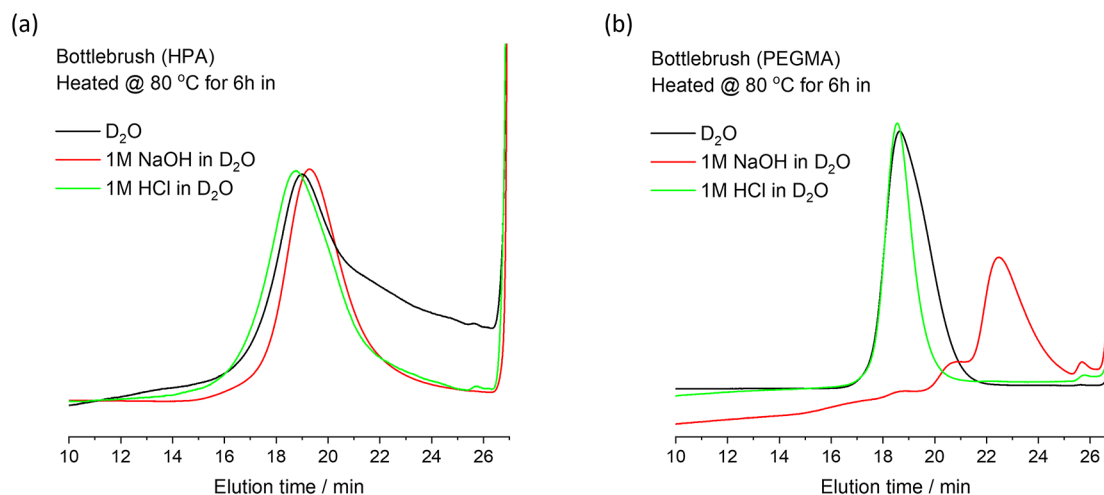
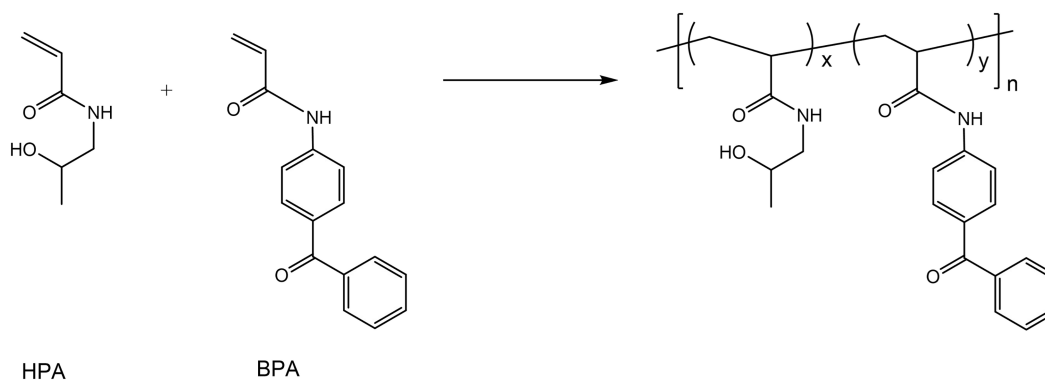
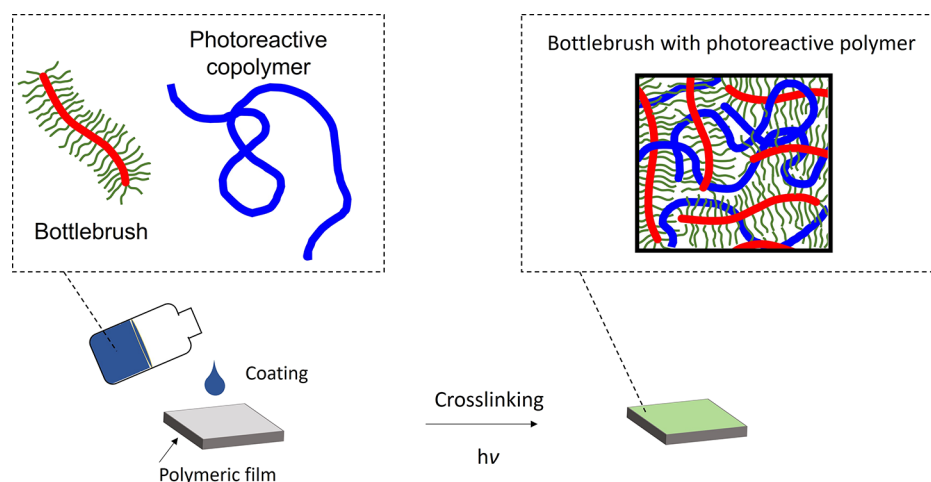
carried out ATRP of HPA from PST(Cl) without adding free initiators. Table 1 shows the characteristics of the obtained bottlebrushes. Since the σ^* was over 0.1, the bottlebrush was confirmed to have a CPB structure.

Stability of the Bottlebrushes against Hydrolysis. To confirm our hypothesis that polyacrylamide polymers and polystyrene derivatives would be more stable against hydrolysis compared with their polymethacrylate analogous polymer counterparts, we carried out a hydrolysis test on the bottlebrushes composed of poly(HPA) as well as poly(PEGMA) (Figure S1). Herein, the degradation of the bottlebrushes in a strong acidic and a basic conditions were monitored by ^1H NMR measurements. Since it is known that hydrolysis can be accelerated by increasing the temperature, we carried out stability tests at 80°C . As shown in Figure S2, the NMR tubes containing the bottlebrush solutions (at pH = 1 or 14) were heated in an oil bath. At 0, 3, and 6 h, the bottlebrush polymers were confirmed by ^1H NMR. Figure S3 shows NMR charts of the bottlebrushes in the aqueous solutions. No peak shifts were observed on the bottlebrush (HPA) after exposure to either acidic or basic conditions. On the other hand, the bottlebrush (PEGMA) showed peak shifts around 3.36 and 3.1 ppm in the basic condition (pH = 14), suggesting a structure change by hydrolysis. To further investigate this, the polymer solutions after 6 h treatment were neutralized and measured by GPC. Figure 3 shows the GPC traces obtained from the bottlebrush polymers. As shown in Figure 3a, the GPC charts of the bottlebrush (HPA) were almost identical independent of the aqueous condition (pH = 1 or 14). This indicates that this bottlebrush polymer was stable under the conditions examined. However, the GPC trace of the bottlebrush (PEGMA) incubated at the basic condition (pH = 14) significantly shifted to a higher elution time (indicating a smaller molecular weight) compared with that in the neutral and acidic conditions. This indicates that the graft chains (PPEGMA) were hydrolyzed at least partially. These results confirmed our hypothesis that the polyacrylamide polymers and polystyrene derivatives are more stable against hydrolysis, especially at basic conditions, compared with their analogous polymethacrylate counterparts.

Preparation of Bottlebrush Coatings. In the previous report, we cross-linked bottlebrushes at graft chain ends by

Table 1. Characteristics of the Macroinitiator and Bottlebrush Polymers Used in This Study

polymers	M_n	M_w/M_n	conversion (%)	σ_{eff} (nm ²)	σ^*
macroinitiator (PST-Cl)	29,600	1.2	30		
bottlebrush	180,000	1.2	27	0.2	0.2

**Figure 3.** GPC traces after heating of bottlebrush polymers in different aqueous solutions at 80 °C for 6 h. (a) GPC charts for HPA bottlebrush and (b) GPC charts for PPEGMA bottlebrush.**Scheme 2. Synthesis of Photoreactive Copolymers (Poly(HPA-*r*-BPA))****Scheme 3. Schematic Illustration of the Coating Procedure**

radical coupling (Scheme S1a). This method can draw the best performance of the bottlebrushes, namely, the CPB effects

such as bioinertness and super-lubrication property;^{13–19} however, the substrates need to have halogens (e.g., Br or

Cl) to covalently immobilize the bottlebrushes, and the physical stability of the cross-linked film is skeptical because radical coupling (cross-linking) does not occur at all graft chain-ends. For example, the coupling efficiency was about 25% for the PS-g-PBIEM (CPB type).³³ Thus, to overcome these problems, we planned to use photoreactive copolymers made from HPA and BPA monomers that we previously developed (Scheme 2).²⁷ As shown in Scheme S1b, the benzophenone groups in the copolymer can generate radicals under UV irradiation, which in turn react with hydrocarbon-based materials or lead to radical coupling of two activated benzophenone groups. This photoreactive copolymer is very useful because it is applicable for all polymeric substrates, and UV treatment is easy and its reaction time is very short (a few minutes by a typical UV lamp). In addition, the cross-linking degree, namely, the physical stability is controllable by adjusting the amount of benzophenone groups in the copolymer. Thus, we plan to incorporate the photoreactive polymer to assist the bottlebrush polymer coatings (Scheme 3).

First, we prepared the photoreactive copolymers with different BPA contents by conventional radical polymerization (Scheme 2). Table 2 summarizes characteristics of the

Table 2. Characteristics of the Different Photoreactive Copolymers Used in This Study

polymers	BPA content	M_n	M_w/M_n
poly(HPA ₉₉ - <i>r</i> -BPA ₁)	1 mol	170,000	1.9
poly(HPA ₉₅ - <i>r</i> -BPA ₅)	5 mol	155,000	1.7
poly(HPA ₉₀ - <i>r</i> -BPA ₁₀)	10 mol	153,000	2.0

copolymers. The values of M_n and M_w/M_n were similar for all three copolymers, and the detected BPA content was identical to the initial feeding monomer ratios. Then, we prepared polymer solutions with the photoreactive copolymers and bottlebrushes as shown in Table 3 and spin coated the solutions with or without bottlebrushes on silicon wafers. Prior to the spin coating, AMP was immobilized on silicon wafers to enhance the bonding between the coating and the silicon wafers. The coating films were then cross-linked by UV irradiation for 4 min, and the dry thicknesses were measured by ellipsometry. We also varied the mixing ratio of the bottlebrush polymers to the photoreactive copolymers from 1.0 to 2.0 (w/w). However, after washing, the coatings in which the photoreactive copolymers with 1 and 5 mol % BPA significantly decreased their thicknesses. The result indicates that the BPA groups were not enough to covalently keep all the

bottlebrushes (data not shown here). Thus, herein, we adopted the mixing ratio of 1 for the photoreactive copolymers with 1 and 5 mol % BPA and 2 for that with 10 mol BPA. We confirmed that the dry thickness changed little after washing (see Table S2), indicating that the cross-linking reaction was successful in keeping the bottlebrush polymers in the coatings as well as forming stable coatings on the substrates. Then, the surface hydrophilicity of the coatings was evaluated by contact angle measurements (Table 3). Interestingly, the surfaces became more hydrophobic upon the addition of the bottlebrush polymers. In addition, XPS analysis was performed on the coatings prepared on PET films in comparison to untreated PET film control substrates. Table S3 shows the elemental composition, obtained from survey spectra, of these surfaces as well as the contributions of chemically different carbon species obtained from high-resolution C 1s spectra. In comparison with the untreated PET film, all the coatings with or without bottlebrushes showed higher concentrations of N originating from the amide functional group associated with the HPA and BPA components. On the other hand, the coatings with bottlebrushes showed increased C5 + C6 + C7 + C8 components with increasing BPA content in the photoreactive copolymer. This was attributed to the aromatic ring present in BPA.

To demonstrate that our coating technology is applicable for a large ship's hull, we also investigated the cross-linking capability of the photoreactive coatings by a ceiling lamp (fluorescent tube lamp) and sunlight (Figure S4). As an example, we used the BPA 5 mol % copolymer. The copolymer was spin-coated on AMP immobilized silicon wafers. Then, the coated samples were placed on the laboratory table (Figure S4a). The distance between the table and the fluorescent tube lamp was ca. 2.2 m. After 24 h, the coating films were thoroughly washed with fresh ethanol, and the thickness was confirmed by an ellipsometer. The thickness changed little, indicating completion of the cross-linking. We also examined 1 h irradiation, but the thickness slightly decreased, indicating that 1 h was not enough to complete cross-linking (Figure S5a). The UV intensity of the ceiling lamp is much lower than that of the UV lamp, and therefore, this result was expected. Next, we placed the coating outside and exposed it to direct sunlight (Figure S4b). Since it is known that sunlight (average 6 mW/cm² in Tsukuba, Japan) has more UV intensity than the ceiling lamp, we expected that the cross-linking would be completed faster than 24 h. In fact, the cross-linking was completed in around 30 min by sunlight (no thickness changes) (Figure S5b). The thickness changes before and after washing are summarized in Table 4. These results prove that

Table 3. Coatings on PET Films Used in This Study and the Static Water Contact Angle Measured on These Coatings

coating sample	HPA/BPA copolymer	mixing ratio	contact angle (°)
	BPA content	copolymer/bottlebrush (wt/wt)	
BPA 1 mol %	1 mol		39.4 ± 0.5
BPA 5 mol %	5 mol		38.4 ± 0.8
BPA 10 mol %	10 mol		40.9 ± 1.3
bottlebrush with BPA 1 mol %	1 mol	1/1 ^a	48.2 ± 0.7
bottlebrush with BPA 5 mol %	5 mol	1/1 ^a	48.1 ± 0.8
bottlebrush with BPA 10 mol %	10 mol	1/2 ^b	50.3 ± 0.6

^aThe ethanol solutions contained 1.5 wt % BPA copolymer and 1.5 wt % bottlebrush polymer. ^bThe ethanol solution contained 1.0 wt % BPA copolymer and 2.0 wt % bottlebrush polymer. The overall polymer content (BPA copolymer + bottlebrush) of the solutions was 3.0 wt % in all cases.

out coating technology does not need a complex process but instead just requires the paint to be applied in direct sunlight.

Table 4. Dry Thickness of the Coatings before and after Washing^a

UV source	irradiation time (h)	before washing (nm)	after washing (nm)
ceiling lamp	0 ^b	218.6 ± 9.0	9.6 ± 0.3
	0.5	229.9 ± 7.9	166.4 ± 9.4
	1	233.4 ± 7.4	185.4 ± 9.8
	24	233.1 ± 6.7	226.8 ± 10.8
sunlight	0 ^b	218.6 ± 9.0	9.6 ± 0.3
	0.5	237.6 ± 2.6	234.8 ± 4.5
	1	247.4 ± 1.0	238.9 ± 3.8
	3	213.1 ± 4.2	211.5 ± 1.0

^aThe coatings were prepared on AMP immobilized silicon wafers. The concentration of the photoreactive polymers (BPA 5 mol %) is 3.0 wt % in ethanol. ^bSame sample.

Long-Term Stability of the Coatings in Artificial Seawater. To confirm the long-term stability of the bottlebrush coatings, we immersed the coatings with or without bottlebrushes prepared on PET films in artificial seawater at 25 °C for up to 3 months. Every 2 or 3 days, the seawater was exchanged. At 1, 2, and 3 months, the coatings were taken out of the seawater, and the dry thicknesses were investigated using an ellipsometer. Figure 4 shows the dry thickness of the coatings observed on PET films. Notably, the thickness changed little on all of the coatings over the 3 month period even though the bottlebrush coatings experienced multiple swelling/deswelling cycles. This result indicates that the cross-linked coatings are stable and that the amide functional groups in the HPA and BPA components as well as the macroinitiator (polystyrene derivative) provide appropriate long-term stability in aqueous solutions.

Bioinertness of the Bottlebrush Coatings. We previously reported that CPB-type bottlebrushes showed excellent nonbiofouling properties compared with SDPB-type bottlebrushes, in which the cross-linked coatings were composed of only bottlebrushes.²⁰ On the other hand, herein, we used photoreactive copolymers to covalently immobilize bottlebrushes on the substrates (Scheme 3). As mentioned above, the incorporation of the photoreactive copolymer into the bottlebrush coatings may diminish bioinertness, one of the CPB effects, depending on the molar ratio of the BPA groups used and/or the mixing ratio of the copolymers. Thus, to confirm this, we carried out mammalian cell attachment tests on the coatings prepared under the conditions listed in Table 3. Herein, we used L929 mouse fibroblast cells, one of the cell lines commonly used for cell attachment tests. In previous works, we used L929 cells to demonstrate bioinertness of the bottlebrush coatings and CPBs.^{20,21,26} Therefore, it is reasonable to confirm the bioinert properties of our coatings using L929 cells after dipping in artificial seawater. We seeded 5×10^4 cells cm^{-2} on the coatings prepared on PET films and a bare PET film as a control. After 24 h incubation, the adherent cells were fixed and stained with fluorescent dyes to count (Figure S5). Figure 5a shows the adherent L929 cells on the surfaces. The results were normalized by using the number of adherent cells on bare PET as 100%. Following our observations in previous reports, BPA 1 mol % suppressed cell adhesion, but BPA 5 and 10 mol % showed some cell adhesion.

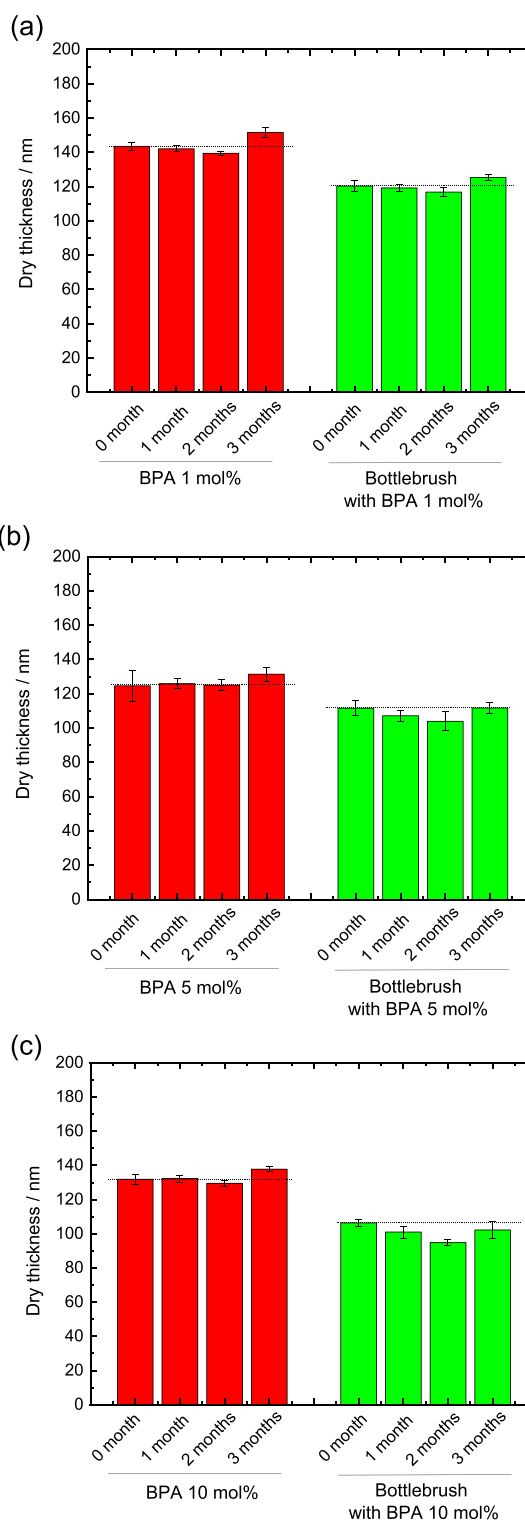


Figure 4. Dry thickness of coatings prepared on PET films before and after dipping in artificial seawater for up to 3 months at 25 °C. "0 month" denotes the pristine coating before dipping in the seawater. Photoreactive polymers with a BPA content of (a) 1 mol %, (b) 5 mol %, and (c) 10 mol % were used.

On the other hand, the photoreactive polymers containing bottlebrush polymers suppressed cell attachment to almost zero independent of the BPA content. This effectiveness in cell attachment prevention was similar to our previous report (in which the coatings were composed of only CPB-type

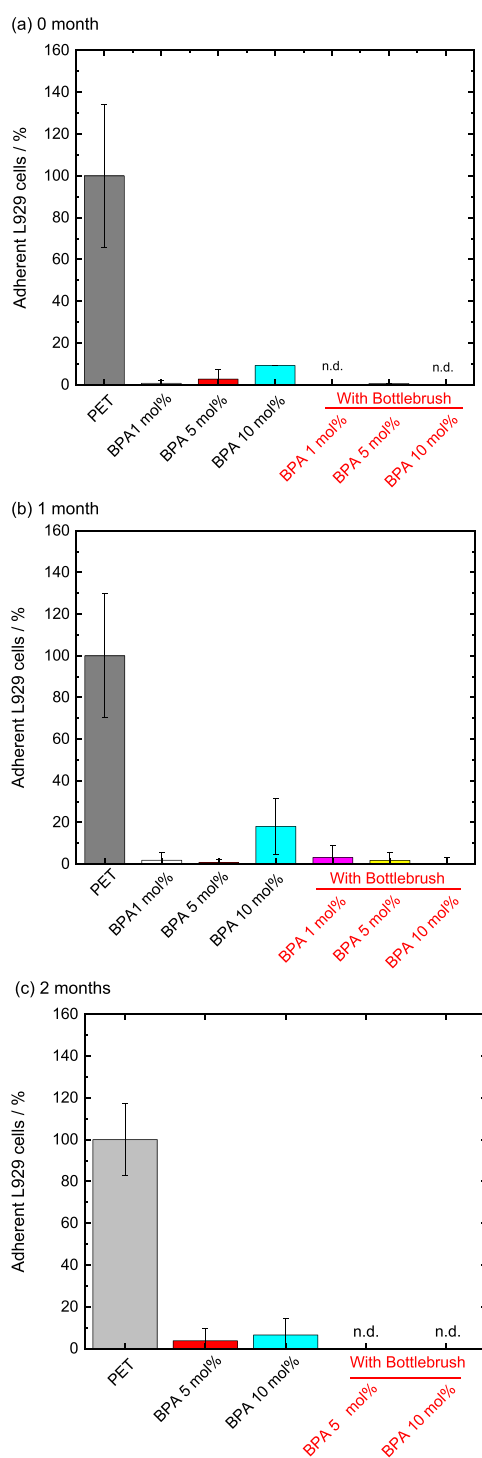


Figure 5. L929 adhesion on PET samples with and without coatings (a) before dipping (denoted as 0 month) and after dipping in artificial seawater for (b) 1 month and (c) 2 months. Coatings containing 1 mol % BPA were not investigated at the 2 month time point. $[L929]_0 = 5 \times 10^4 \text{ cells cm}^{-2}$. Incubation time = 24 h. The y axis is normalized to the number of adherent cells detected on fresh PET samples.

bottlebrushes).²⁰ This result strongly suggests that the coatings produced in this study provide sufficient amounts of bottlebrushes on their surface to effectively prevent biofouling.

To examine the long-term nonbiofouling properties of the coatings, we also examined the attachment of L929 cells on coatings that were immersed in seawater for 1 and 2 months. If

seawater damages the coating surfaces, e.g., by chemical decomposition, the bioinert properties of the coatings should be impaired. Figure 5b shows the adherent L929 cells on the coatings immersed in the seawater for a month. Moreover, to confirm whether seawater does not damage the PET substrate, we also immersed the bare PET film in seawater for a month (denoted as PET). No significant difference was observed between PET and fresh PET films, indicating that the seawater did not affect the PET. After 1 month immersion, coatings also yielded similar results to those before immersion. Figure 5c shows the cell attachment results on the coatings after 2 month immersion. Since coatings are expected to be more stable with increasing BPA content in the copolymer, we only used the copolymers with 5 and 10 mol % BPA for the 2 month immersion experiment. All coatings with or without bottlebrushes show the same trend in cell attachment as observed for the 0 and 1 month immersion coatings. The results confirm that the coatings are physically and chemically stable in artificial seawater over several months and that they maintain their bioinertness over this period.

Barnacle Cypris Larva Settlement Testing. Marine biofouling, which is caused by the settlement of marine organisms such as seaweeds and barnacles on submerged surfaces, is detrimental to a broad range of marine applications. Due to the importance of this problem, in vitro tests have been developed to examine surfaces and materials in regard to their biofouling behavior. Here, we carried out a barnacle cypris larva settlement test on the coatings. Based on the results from the L929 cell adhesion test, we chose the bottlebrush coatings using BPA 5 mol % as the best candidate. Herein, the cross-linked coatings were prepared on PVC films because cypris larvae adhere well on bare PVC films compared with PET films, and hence, these are suitable as positive controls. About 100 larvae were seeded on the coatings as well as untreated PVC films and glass control samples (Figure S7). The larvae were kept at room temperature and monitored for 9 days. Settlement behaviors of cypris larvae are illustrated in Scheme S2: (1) Larvae swim in water using their thoracic appendages, and then (2) the larvae begin to walk on substrate materials using their antennules in search of a suitable area to cement themselves. After this exploration phase, (3) the larvae metamorphose into juveniles. (4) Subsequently, the firmly adhered juveniles metamorphose into young barnacles. On the other hand, (5) damaged or dying larvae lie down on the substrate with some reaction such as moving thoracic appendages or (6) sticking out thoracic appendages without any reaction. We counted larvae in phases (3) and (4) as "settlement" (Table S4). Figure 6a shows the time dependence of settled larvae on the surfaces. Larvae started to settle on glass at 3 days and on bare PVC at 5 days. In contrast, the coatings with or without bottlebrush much suppressed the larva settlement. Figure 6b shows the number of settled larvae on the surface at 9 days. Figure S8 shows pictures of the settled larvae on the surfaces. Firmly settled larvae were observed on glass and PVC films. Almost all larvae settled on the glass, and approx. 50% of larvae settled on the PVC film. While BPA 5 mol % had some larva settlement ($2.25 \pm 0.6\%$), the bottlebrush coatings perfectly prevented larva settlement (0%, not detected). This result strongly suggests that the bottlebrush coatings would be an effective surface for the prevention of marine organisms.

Static Immersion Test in the Sea (Field Test). To further investigate the antifouling properties of the bottlebrush

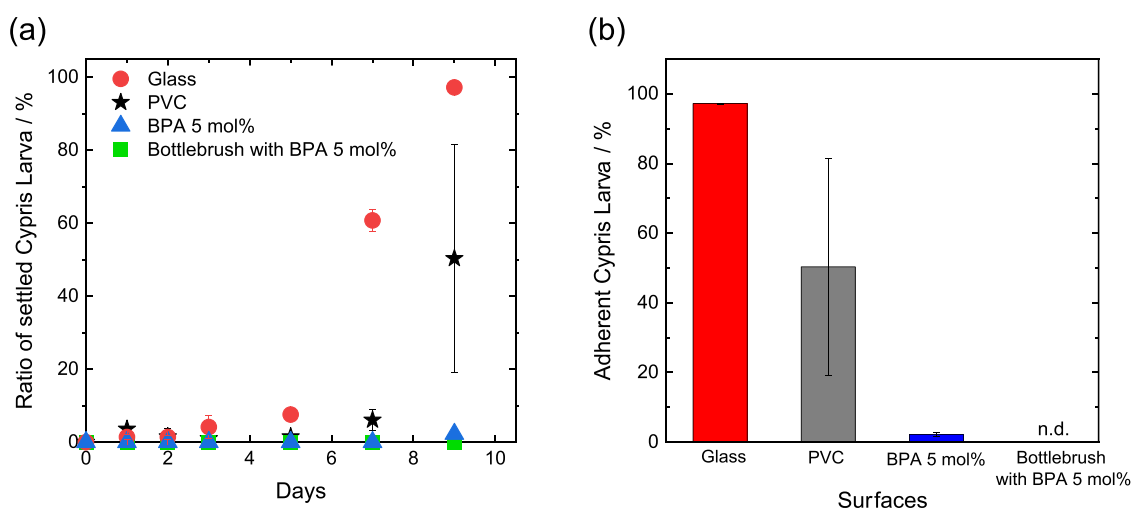


Figure 6. (a) Time dependence of settled larvae on the surfaces. (b) The number of settled larvae on the surface at 9 days. Mean value $\pm$ standard deviation ($n = 2$). $[\text{Cypris larvae}]_0 = \sim 100$ individuals/surface. The y axis is normalized to the initial number of seeded cypris larvae.

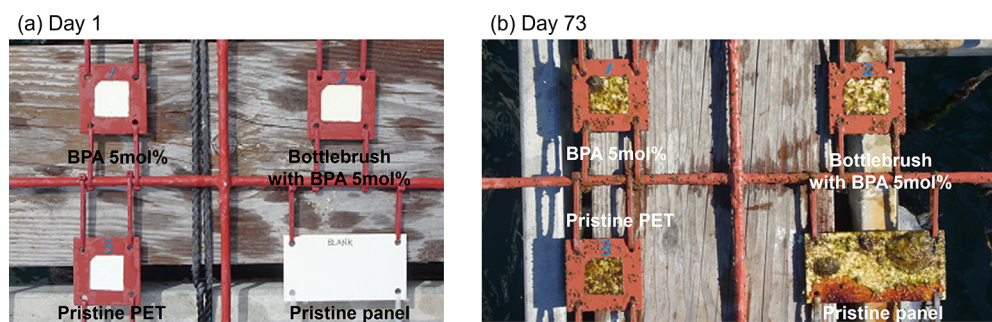


Figure 7. Pictures of the test panels (polymer coatings and controls) for (a) 1 day immersion and (b) 73 day immersion.

coatings, we conducted a field test. The BPA 5 mol % coatings with or without bottlebrush were prepared on PET films, and test panels having the coated films as well as an untreated PET film were immersed in the sea for 2.4 months (73 days). The panels were taken out of the sea after 1 and 73 days, respectively, and settled marine organisms were observed. Pictures are shown in Figure 7. In Figure 7b, all surfaces were more or less covered with a brown substance. This observation is due to the attachment of diatoms that belong to the group of unicellular algae in unequal-haired plants. Different from other marine organisms, they do not provide significant resistance and are easily removed when a marine vessel is moving through water. Therefore, these are usually not counted as fouling. In the case of static soaking tests, it is common to see diatoms even on commercially available antifouling paints that contain biocides. Thus, we did not consider adhered diatoms as fouling. The marine organisms settled on the surfaces are listed in Table 5. The fouling area covered with the listed organisms is also shown in the table. At day 1, settlement of marine organisms was almost absent on all the surfaces. However, after 73 days, the BPA 5 mol % had barnacle settlement and the pristine PET film had Serpulidae settlement. On the other hand, the bottlebrush coating did not have any settlement, demonstrating excellent antifouling properties (fouling area = 0%). This result agreed well with the long-term stability test in artificial seawater and the cypris larva settlement tests. Due to their long-term stability, effective antifouling properties, environmentally friendly (biocide-free) nature, and ease of manufacturing, the bottlebrush coatings

using photoreactive copolymers are expected to be developed further into specialty marine paints. Here, the transparent nature of the coating may also present a significant advantage in comparison to existing alternatives.

CONCLUSIONS

We developed a next-generation, nontoxic marine antibiofouling coating method that can be applied in one step, provides excellent biostability, and provides an environmentally sustainable alternative to current antibiofouling coatings that are based on the continuous release of toxins. The strong resistance against acid and base hydrolysis of bottlebrush HPA polymers was demonstrated in the accelerated degradation experiments at 80 °C. Moreover, the coating thickness was found to remain stable after incubation for 3 months in sea water. The bottlebrush coatings assisted by the 5 mol % BPA copolymer prevented cypris larva attachment for up to 9 days and prevented the settling of marine organisms in the sea for up to 73 days. In addition, we demonstrated that cross-linking of the photoreactive copolymer was completed after 30 min exposure to sunlight. Due to the ease of the coating process, long-term durability, and effective antifouling performance, our bottlebrush coating technology is expected to be exploited in next-generation, biocide-free marine paints. We are now confirming that industrial coating techniques such as a bar-coating, spray-coating, and brush-coating can be used to apply our bottlebrush coatings. Following on from this, we are planning to conduct a feasibility study with the bottlebrush coatings on a section of a marine vessel.

Table 5. Settled Marine Organisms on the Test Panels^a and the Ratio of Fouling Area to Substrates (5 × 5 cm²): (a) 1 Day Immersion and (b) 73 Day Immersion

(a) Day 1.

Sample	BPA 5%	Bottlebrush with BPA 5%	Pristine PET	Pristine panel
Fouling area / %	0	0	0	NA ^b
Barnacle				
Serpulidae	✓			
Conch				
Mytiliidae				
Colonial sea squirts				
Styela plicata				
Ciona savignyi				
Bryozoa				
Porifera				
Green algae				

(b) Day 73.

Sample	BPA 5%	Bottlebrush with BPA 5%	Pristine PET	Pristine panel
Fouling area / %	1 ~ 5	0	1 ~ 5	NA ^b
Barnacle	✓			✓
Serpulidae			✓	✓
Conch				✓
Mytiliidae				
Colonial sea squirts				
Styela plicata				
Ciona savignyi				
Bryozoa				✓
Porifera				
Green algae				

^aBlank box indicates that the organism was not detected. ^bFouling area was not counted on the pristine panel.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.2c06647>.

Structures of (a) PBIEM (macroinitiator) and (b) bottlebrush polymer (Figure S1); schematic of the experimental setup (Figure S2); hydrolysis of bottlebrushes (Figure S3); pictures of the photoreactive polymers (Figure S4) and photoreactive coatings (Figure S5); fluorescent images of adherent L929 mouse fibroblast cells (Figure S6); picture of the experimental setup of the cypris larva settlement test (Figure S7); images of settled cypris larvae (Figure S8); reaction scheme of the radical coupling (a) of chain-end halogen of bottlebrushes and (b) of benzophenone groups (Scheme S1); behaviors and actions of cypris larvae during the settlement test (Scheme S2); characteristics of bottlebrush polymers (Table S1); dry thickness of the coatings (Table S2); XPS C 1s high-resolution data (Table S3); and cypris larva behavior and actions (Table S4) (PDF)

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Notes

The authors declare no competing financial interest.

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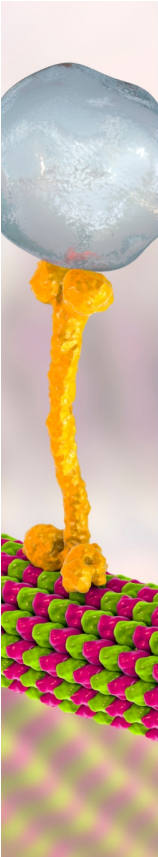
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